



JEAN L. PIE

NEUROSCIENTIST | IN VIVO ELECTROPHYSIOLOGIST

SKILLS

- Data Analysis and Statistics:
Python, MATLAB, R, Git
- Setting up laboratoria:
Obtaining permits,
Implementing Government
and Ethical Guidelines, Data
acquisition, Arduino C+,
Electrical Engineering.
- Public Speaking: Prizes for
Best Pitch (ONWAR Graduate
School Meeting 2019), Best
Story Slam (Amsterdam
Neuroscience Event 2023)
- Collaborate in International
Environments
- Problem Solving
- Collaboration
- Teaching

LANGUAGES

- English (Fluent)
- French (Native)
- Spanish (Native)
- Dutch (B1+)

CONTACT

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- 📍 Amsterdam
- 🔗 gitlab.com/jeanlpie

WORK EXPERIENCE

- **VU Amsterdam, NL** 2021 - 2026
Post-doc, Project Lead
PI: Chris de Kock
 - Lead research project on cell-type specific visuotactile coding in the cortex.
 - Fully setup, collected, cleaned and analysed in-vivo neuronal data (spiking data, LFP, optogenetic tagging of cell-types) and complex behavioral data in behaving mice in a visuotactile VR, with optogenetics.
 - Built data analysis pipeline and coordinated interdisciplinary 3-person team with AI department.
 - Used Machine Learning (single-cell and population decoding: SVM, RF, AUROC), Dimensionality reduction (PCA, Tensor Decomp., subspace projection), and advanced statistical methods (GLM, Linear Mixed Effects, Hierarchical Bootstrapping, custom Permutation Tests, Model Comparison).
 - Daily supervision of 2 PhD students and 5 master interns.
 - Gave Master lecture on neuronal decoding and multisensory integration.
 - Co-authoring 2 publications.
- **Universiteit van Amsterdam, NL** 2017 - 2021
PhD on Visual-Tactile Perception
PI: Cyriel Pennartz
 - Fully set up, collected and analysed spiking data from silicon probes in behaving mice with optogenetic silencing.
 - Published 4 scientific articles, collaborated across two institutions.
 - Taught master courses and daily supervised 3 master interns.
 - Presented work in international conferences and to broad audiences.
- **Biozentrum Basel, Switzerland** 2016
Master Internship
PI: Thomas Mrsic-Flogel
 - Setup optogenetic silencing during context-dependent mouse behavior.
- **Cold Spring Harbor Laboratory, USA** 2014-2015
Internship on Orbitofrontal Cortex study
PI: Adam Kepecs
 - Collected and analysed behavioral and neuronal data. Projection-specific optogenetic identification of neurons during decision-confidence task.
 - Implanted rats with tetrode drives
- **G-LAB - EPFL, Switzerland** 2013
Bachelor Internship
PI: Grégoire Courtine
 - Assisted with experiments: rats on treadmill with closed-loop BCI for spinal cord injury recovery. Motor cortex decoding into spinal cord stimulation.
 - Gait analysis during recovery.

EDUCATION

- **PhD in Neuroscience** 2017 - 2025
"At the Confluence of the Senses: cortical processing of multisensory stimuli and context"
University of Amsterdam, NL
- **MSc in Neuroscience** 2012 - 2016
Ecole Normale Supérieure de Cachan, France
- **BSc Classe Prépa (CPGE) in Biology, Chemistry, Physics and Earth Sciences** 2010 - 2012
Lycée Henri IV, France
National Rank: 40 / ~2000

PUBLICATIONS

- **Pie JL**, van den Heuvel I.K.M., Solovjova S, Mansvelder HD, de Kock CPJ.
Touch responses in S1 layer 5 pyramidal neurons are shaped by multisensory context.
Manuscript submitted. 2026
- **Pie JL**, de Kock CPJ, Olcese U, Pennartz CMA.
Parietal cortex boosts behavioral performance during multisensory localization.
Manuscript submitted. 2026
- **Pie JL**. (PhD Thesis)
At the Confluence of the Senses: Cortical Processing of Multisensory Stimuli and Context.
Jean L. Pie. 2025. ISBN: 9789465340104.
<https://hdl.handle.net/11245.1/f8e75240-c208-4984-b426-4b6eda415acd>
- Mertens EJ, Leibner Y, **Pie JL**, Galakhova AA, Waleboer F, ..., Mansvelder HD, Segev I, de Kock CPJ.
Morpho-electric diversity of human hippocampal CA1 pyramidal neurons.
Cell Rep. 2024 Apr 23;43(4):114100. doi: 10.1016/j.celrep.2024.114100.
- *Shared first authorship:
Oude Lohuis MN*, **Pie JL***, Marchesi P, Montijn JS, de Kock CPJ, Pennartz CMA, Olcese U.
Multisensory task demands temporally extend the causal requirement for visual cortex in perception.
Nat Commun. 2022 May 23;13(1):2864. doi: 10.1038/s41467-022-30600-4.
- de Kock CPJ, **Pie JL**, Pieneman AW, Mease RA, ..., Oberlaender M, Mansvelder HD, Sakmann B.
High-frequency burst spiking in layer 5 thick-tufted pyramids of rat primary somatosensory cortex encodes exploratory touch.
Commun Biol. 2021 Jun 10;4(1):709. doi: 10.1038/s42003-021-02241-8.
- Meijer GT, **Pie JL**, Dolman TL, Pennartz CMA, Lansink CS.
Audiovisual Integration Enhances Stimulus Detection Performance in Mice.
Front Behav Neurosci. 2018 Oct 4;12:231. doi: 10.3389/fnbeh.2018.00231.

PRESENTATIONS

- **Pie JL**. (2023, Dec 08). When is the visual cortex needed for visual perception? [PowerPoint slides].
Amsterdam Neuroscience Event 2023. Prize for Best Story Slam.